2.2 The Gradient

THE ONE SENTENCE

Ask 2.1's question once per knob, stack the answers into one arrow — and that arrow points straight uphill.

Two dials on a real photograph

```
A  = washed-out photo (97,200 numbers)
y  = the correct photo
w₀ = contrast    w₁ = brightness

loss = mean( (w₀·A + w₁ − y)² )
```

1. Wiggle each dial separately

turn CONTRAST a hair → −0.11762

turn BRIGHTNESS a hair → −0.14436

stack: $g = [-0.11762, -0.14436]$

length $|g| = 0.18621$ ← how steep it is here

finite differences, the exact formula and autograd all agree

2. Built one real pixel at a time

is	should be	off	2×A×miss	running
0.700	0.922	−0.221	−0.3101	−0.3101
0.634	0.780	−0.147	−0.1860	−0.2480
0.730	0.973	−0.243	−0.3545	−0.2835
0.639	0.808	−0.168	−0.2154	−0.2665

...all 97,200 numbers average to −0.11762

3. The race — and why nothing can win

5,000 random directions vs the gradient.

Instantaneously: 0 of 5,000 beat it.

$$g \cdot \hat{d} = |g| \cos\theta$$

angle off	score
0°	+0.18621 ← the max
30°	+0.16127
60°	+0.09311
90°	0

Cosine never exceeds 1, and equals 1 in exactly one direction. That is the proof.

⚠ Over a *finite step* a few random directions can win — the surface curves away. Steepest means steepest instantaneously. That gap is 2.3.

4. Perpendicular to the contour

```
g · tangent = 0.00e+00
angle = 90.0000°
```

Exactly, not approximately. Walk along a contour and the score does not change — so the direction of fastest change must be at a right angle to it.

5. Nobody computes the real one

pixels	angle off	downhill	cheaper
1	11.5°	65.7%	97,200×
10	3.8°	97.3%	9,720×
100	1.1°	100%	972×
1000	0.4°	100%	97×

Throw away 99.9% of your data and you still know which way to go. That is what *stochastic / mini-batch / batch* mean.

6. What one gradient costs

7. Advanced

```
GPT-2: 124,439,808 × 4 bytes
     = 498 MB per step
× 300,000 steps = ~149 TB
computed and thrown away
```

- The gradient is local. It is only true where you stand; the surface curves away as soon as you move.
- Its length foreshadows 2.4: an arrow that shrinks toward zero on the way back through layers is the vanishing gradient.
- No gradient at all: the moment a model *picks* a token, a black-box API, a human preference. Those need a different toolbox entirely.

NEXT: you know which way. How far? → 2.3, the learning rate.